ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** particles move faster at higher temperature	Higher kinetic energy.
2	**(a)** gives out extra heat when it condenses on the skin	Condensation of steam releases the heat absorbed during vaporisation.
3	**(d)** liquid	Liquefaction.
4	**(c)** particles of salt fit in the gaps between water particles	Spaces between particles.
5	**(b)** boiling point (condensation point)	Condensation.
6	**(a)** evaporates quickly	Takes heat.
7	**(c)** solid, liquid, gas	Solids have the least gaps.
8	**(a)** water at 90 degrees Celsius	Higher temperature.
9	**(b)** vapour cannot escape; condensation balances evaporation	Equilibrium.
10	**(a)** diffusion in air takes time	Particles travel gradually.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Particles of one fit into the spaces of the other (1); This shows that there are gaps between particles (1).	1+1
12	Higher temperature increases particle speed (1); Diffusion becomes faster (1).	1+1
13	Vapour is the gaseous form of a substance that is normally liquid at room temperature (1); Gas is a substance in gaseous state at room temperature, e.g. oxygen (1).	1+1
14	Attraction between water particles pulls them together (1); Forms a small sphere (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Solids: strong forces hold particles (1); Liquids: particles slide over each other (1); Gases: particles fly apart (1).	1+1+1
16	Vapour: particles far apart (1); Condenses: closer, slide (1); Freezes: fixed in a pattern (1).	1+1+1
17	Larger area exposes more particles (1); More evaporate (1); A plate of water dries faster than a glass (1).	1+1+1
	OR: Large gaps in gases (1); Gas particles move fast and freely (1); Fewer collisions slow movement (1).	1+1+1
18	Air is already full of water vapour (1); Less evaporation of sweat (1); Less cooling effect (1).	1+1+1
	OR: Melting is change of state by heat (1); Dissolving is mixing a solute into a solvent (1); Dissolving needs a solvent (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Place a crystal in water (1); Colour spreads without stirring (1); Motion of particles (1); Spaces: dissolves with little volume change (1); Faster in warm water (1).	1+1+1+1+1
	OR: Compress the gas (1); Cool it (1); LPG (1); CNG and liquid oxygen (1); Liquid nitrogen (1).	1+1+1+1+1
20	-10 to 0: ice warms (1); At 0: melting (1); 0 to 100: water warms (1); At 100: boiling (1); Above 100: steam warms (1).	1+1+1+1+1
	OR: Saucer, bowl, glass (1); Larger surface dries faster (1); More particles escape (1); Same temperature and wind (1); Surface area matters (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that particles of a solid are at rest; they vibrate about fixed positions.
- Mixing up the spacing: gas particles are far apart, liquid and solid particles are close.
- Saying temperature keeps rising during melting or boiling; it stays constant during a change of state.
- Writing that evaporation happens only at the boiling point, or that it heats the surroundings; it causes cooling.
- Confusing diffusion (mixing of particles by themselves) with dissolving or with stirring.

- Stating that a liquid has a definite shape; it has a definite volume only.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** evaporation	Liquid to vapour.
2	**(a)** increased pressure raises the boiling point	High pressure.
3	**(c)** increases	More collisions.
4	**(b)** sublimation	Direct solid to gas.
5	**(a)** atmospheric pressure is lower	Low pressure.
6	**(b)** the extra heat is used to change water into steam	Temperature constant during change.
7	**(b)** solid state	Both keep a fixed shape and volume.
8	**(c)** steam at 100 degrees Celsius	Highest temperature.
9	**(c)** forces between particles of X are weaker	Weaker forces allow escape.
10	**(d)** dry, windy place	Fast evaporation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Outside pressure is lower (1); Gas inside expands (1).	1+1
12	Vapour touches the cold surface (1); Condenses to liquid (1).	1+1
13	Particles move faster at a higher temperature (1); Diffusion is faster (1).	1+1
14	Steam contains more heat energy (1); It cooks faster (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tiny pores allow water to seep (1); Water evaporates from the surface (1); Evaporation takes heat and cools the water (1).	1+1+1
16	B has stronger forces (1); More energy is needed to separate particles (1); Hence higher boiling point (1).	1+1+1
17	Heat removed comes from the energy released as the liquid becomes solid (1); The particles settle into fixed positions and give out energy (1); Temperature falls only after freezing is complete (1).	1+1+1
	OR: Steam particles are far apart (1); Steam holds extra heat absorbed during vaporisation (1); Gives severe burns (1).	1+1+1
18	Camphor sublimes slowly (1); Particles escape as vapour (1); Vapour may redeposit on the cool glass (1).	1+1+1
	OR: The liquid in the can evaporates on release (1); Takes heat from the can (1); Cooling results (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Heat is absorbed without a change in temperature during a change of state (1); Melting: heat overcomes the forces holding particles in fixed positions (1); Boiling: heat separates the particles into a gas (1); Steam burns: heat is given out on condensing (1); Ice cools a drink by absorbing heat to melt (1).	1+1+1+1+1
	OR: Diffusion: motion and gaps (1); Pressure: collisions (1); Evaporation: fast particles leave (1); Sublimation: direct escape (1); All use motion and forces (1).	1+1+1+1+1
20	Ice melts by absorbing heat (1); This heat is taken from the drink (1); Water at 0 degrees Celsius only absorbs heat by warming (1); Ice cools more (1); Used in coolers (1).	1+1+1+1+1
	OR: Higher temperature increases rate (1); Wind increases rate (1); Humidity decreases rate (1); Surface increases rate (1); Fair test varying one factor (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that particles of a solid are at rest; they vibrate about fixed positions.
- Mixing up the spacing: gas particles are far apart, liquid and solid particles are close.
- Saying temperature keeps rising during melting or boiling; it stays constant during a change of state.

- Writing that evaporation happens only at the boiling point, or that it heats the surroundings; it causes cooling.
- Confusing diffusion (mixing of particles by themselves) with dissolving or with stirring.
- Stating that a liquid has a definite shape; it has a definite volume only.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** particles are held in fixed positions by strong forces	Strong forces.
2	**(b)** hot and windy day	More evaporation.
3	**(c)** gets slightly warmed by the vapour	Latent heat released.
4	**(c)** particles slide over each other but stay close	Intermediate forces.
5	**(a)** temperature rises to 0 degrees Celsius	It must warm first.
6	**(d)** equilibrium and coexist	Both present.
7	**(a)** ice absorbs heat from it to melt	Latent heat of fusion.
8	**(a)** stays at 0 degrees Celsius	Ice melts using the heat.
9	**(a)** are held in fixed positions	They are not fixed.
10	**(b)** a drop of water holds its shape on a leaf	Cohesion.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It absorbs heat as it melts (1); Keeps fish cold (1).	1+1
12	It stays at 80 degrees Celsius (1); Heat is used to change state (1).	1+1
13	Large surface area gives faster evaporation (1); They dry sooner (1).	1+1
14	Water evaporates (1); Particles escape into the air (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lower atmospheric pressure at height lowers boiling point (1); High pressure inside the cooker raises it (1); Boiling point depends on pressure (1).	1+1+1
16	Evaporation from oceans (1); Condensation to form clouds (1); Precipitation as rain or snow (1).	1+1+1
17	Sun's heat evaporates sea water (1); Water vapour rises (1); Forms clouds (1).	1+1+1
	OR: Wet cotton wool wrapped on a thermometer bulb (1); Temperature falls as water evaporates (1); Use acetone to show larger fall (1).	1+1+1
18	Particles move faster (1); Collide with walls more often and harder (1); Pressure increases (1).	1+1+1
	OR: Gas particles are far apart (1); Few particles in the same volume (1); Hence low mass per volume (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Temperature (1); Place (1); Speed (1); Bubbles (1); Example (1).	1+1+1+1+1
	OR: More evaporation from the sea (1); Air holds more vapour (1); Sweat dries slowly (1); Clothes dry slowly (1); Feel sticky (1).	1+1+1+1+1
20	Solid: fixed positions (1); Liquid: slide (1); Gas: free (1); Compressibility differ (1); Flow differs (1).	1+1+1+1+1
	OR: Solid: vibration in fixed positions (1); Heated: vibration increases (1); Melted: pattern breaks, particles slide (1); Boiled: particles escape (1); Energy changes arrangement (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that particles of a solid are at rest; they vibrate about fixed positions.
- Mixing up the spacing: gas particles are far apart, liquid and solid particles are close.
- Saying temperature keeps rising during melting or boiling; it stays constant during a change of state.
- Writing that evaporation happens only at the boiling point, or that it heats the surroundings; it causes cooling.
- Confusing diffusion (mixing of particles by themselves) with dissolving or with stirring.
- Stating that a liquid has a definite shape; it has a definite volume only.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND